

Homework 1

(Chapter 2 Vacuum, total points: 100)

- (1) (i) Please derive the Knudsen equation from $J_i = \frac{1}{4} n \bar{v}$. (ii) An electron-beam evaporator is used to deposit Ti metal at a deposition rate of 0.05 nm s⁻¹. The atomic weight and density of Ti metal are 47.867 g mol⁻¹ and 4.502 g cm⁻³. Assume that all Ti atoms flying to the substrate surface are adhered to the surface. Please calculate the atomic impingement flux of Ti. (5 + 5 = 10 points).
- (2) Please derive the expression for the most probable velocity v_{mp} of gas molecules according to the Maxwell-Boltzmann distribution. Please provide the detailed derivation steps instead of simply giving the result, because one can easily find the answer on the web. According to the obtained expression, calculate the most probable velocity of oxygen (O₂) gas molecules at room temperature (25 °C). Please also calculate the average velocity and root-mean-square velocity of oxygen (O₂) gas molecules at room temperature (25 °C) and order them from small to large (8 + 2 + 2 + 2 + 1 = 15 points).

Homework 1

(Chapter 2 Vacuum, total points: 100)

- (3) A 1.0 m^3 cubic chamber contains O_2 molecules at a pressure of 10^{-3} atm at 300 K . How many molecules are there in the chamber? What fractions of gas molecules in this chamber have translational kinetic energies larger than $1k_{\text{B}}T$ and $10k_{\text{B}}T$, respectively. You will need to use some software to help you to find out definite integrals ($5 + 5 + 5 = 15$ points).
- (4) A closed chamber is filled with N_2 . The chamber is maintained at a temperature of 25°C and a pressure of 10 mPa (high vacuum). The diameter of N_2 molecules is 0.3 nm . Please calculate the mean free path of the N_2 molecules in the chamber and average spacing between two N_2 molecules (assume each N_2 molecule occupies a cubic volume). If the pressure is increased to the atmosphere pressure, please calculate the mean free path and the average spacing again ($5 + 5 + 5 + 5 = 20$ points).

Homework 1

(Chapter 2 Vacuum, total points: 100)

- (5) Gold films are often used in scientific experiments. They are usually deposited at a rate of 0.1 nm s^{-1} by electron-beam evaporation. The density of gold is 19.30 g cm^{-3} . Assume that the gold vapor is made of gold atoms. Please calculate the deposition flux. The temperature of the chamber is 50°C . Assume that the sticking coefficient of gold on the substrate is 1.0. Please determine the partial pressure of the gold vapor during deposition (10 + 10 = 20 points).
- (6) A process chamber of 1.0 m^2 surface area is outgassing at a rate of $10^{13} \text{ molecules cm}^{-2} \text{ s}^{-1}$. For a process-gas flow of 10 sccm and a total pressure of 50 Pa, what are the molecular percentage and partial pressure of the outgassing impurities? For a process-gas flow of 100 sccm and a total pressure of 100 Pa, what are the molecular percentage and partial pressure of the outgassing impurities (5 + 5 + 5 + 5 = 20 points)?

Homework 1

(Chapter 2 Vacuum, total points: 100)

Please submit your solutions by 11:30 pm on 23 September 2026 (Wednesday) in Blackboard. In case you have difficulty in uploading your solutions on Blackboard, please send your solutions by email in time to Ms. DING Yan (yding@phy.cuhk.edu.hk) or Mr. ZHAO Keyuan (kyzhao@phy.cuhk.edu.hk).

Late submission will cause points to be reduced according to how late a submission is.